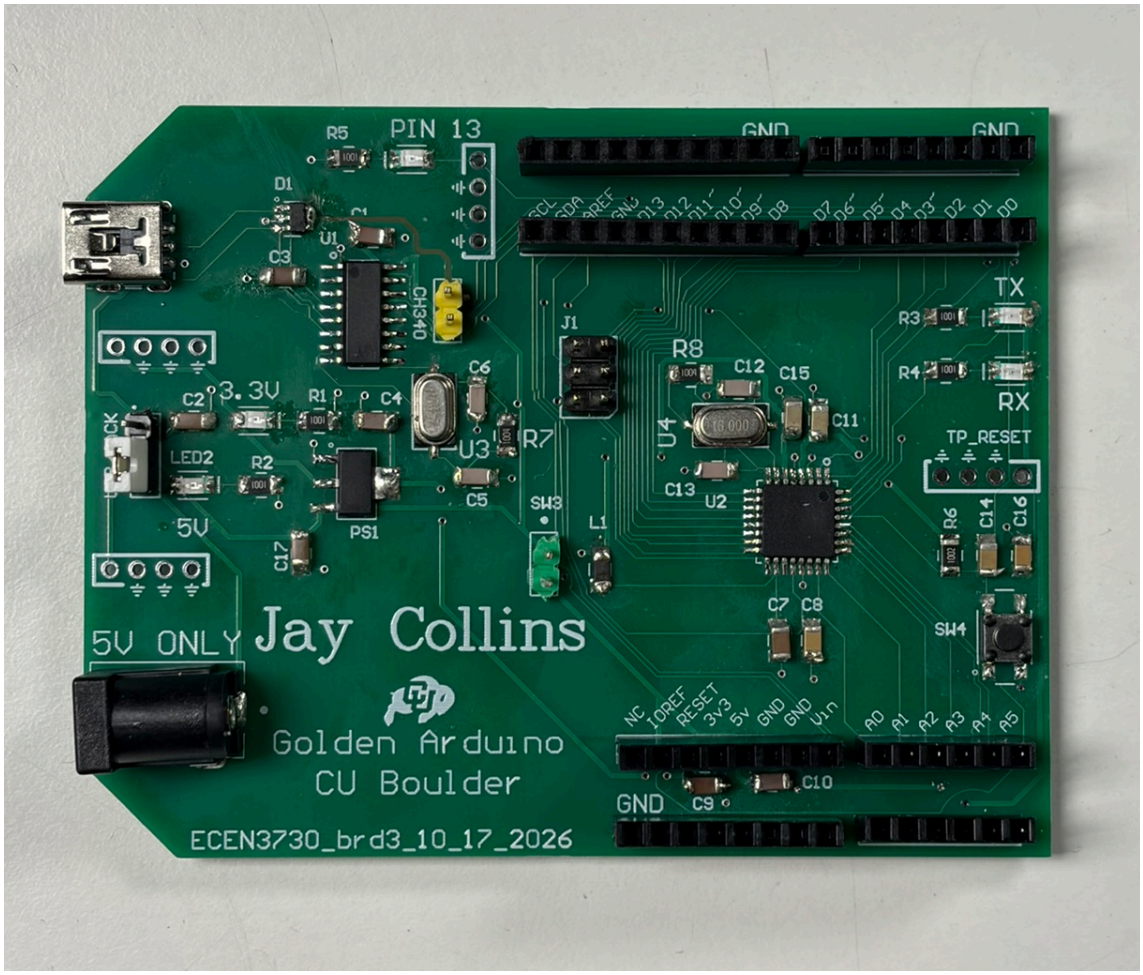
ECEN 3730: Board 3 Report

Golden Arduino Board



Jay Collins

Description:

This board is a Golden Arduino Uno. It uses the same ATmega328P microcontroller, CH340G USB to UART chip and header layout as a commercial Arduino Uno so that any shield fits, but the layout follows best design practices to reduce switching noise. It is powered from either a USB mini connector or a 5V charger through a switch, and it has isolation switches, indicator LEDs and test points for debugging. I measured its switching noise and near field emissions with the noise shield and compared them to a commercial Whadda Arduino Uno board.

Plan of Record:

The circuit board contains:

1. A 2 layer board under 3.9 x 3.9 inches, with a continuous ground plane on the bottom and no copper pour on the top
2. An ATmega328P (TQFP-32) with a 16 MHz crystal, 22 pF capacitors and a 1 MΩ feedback resistor
3. A CH340G USB to UART chip powered from 5V, with a 12 MHz crystal, 22 pF capacitors and a 1 MΩ feedback resistor. Its TX and RX cross to the RX and TX of the 328
4. A reset button with a 10k pull-up and a 47 nF capacitor to ground, and a 1 uF series capacitor from the CH340G DTR pin so the computer can reset the 328
5. A USB mini connector and a 5V jack with a slide switch so only one can power the board, and a 3.3V LDO that only feeds its header pin
6. A TVS chip on the USB D+ and D- lines
7. Header sockets in the exact Arduino Uno locations, a second row of ground pins 300 mils outside the digital I/O, and an ICSP connector for burning the bootloader
8. A decoupling capacitor at every VCC pin, a 10 uH ferrite and 22 uF capacitor filter on AVCC, a 47 nF capacitor on AREF, and 22 uF capacitors at the 5V and 3.3V header pins

9. Isolation switches for the CH340G and the 328, indicator LEDs for 5V, 3.3V, TX, RX and pin 13, and test points
10. 1206 passives, 20 mil power traces, labels on every part, and my name and the board revision on the silkscreen

Datasheets:

ATmega328P:

https://www.digikey.com/product-detail/en/microchip-technology/ATMEGA328P-ANR/ATMEGA328PANRCT-ND/2774230

CH340G: https://static.efetividade.net/img/ch340g-datasheet-34852.pdf

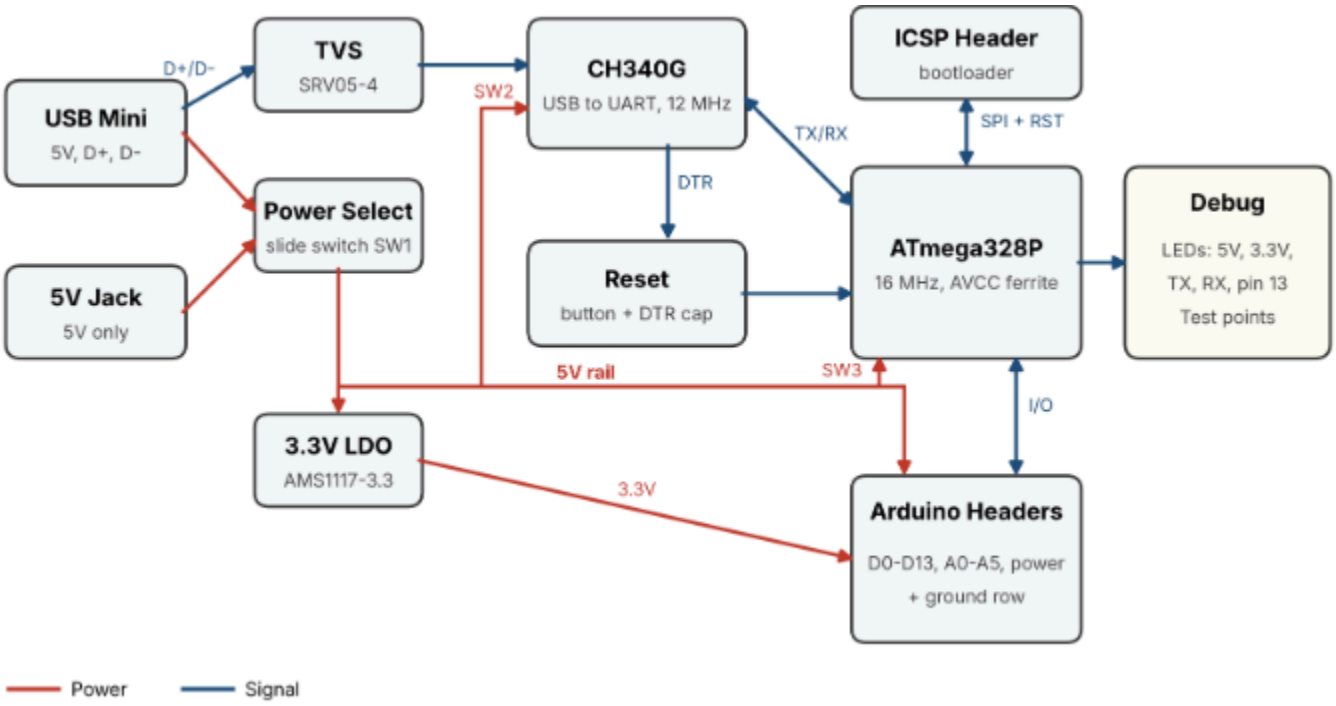
TVS chip (Protek SRV05-4-P-T7):

https://www.lcsc.com/product-detail/TVS_SRV05-4-P-T7_C85364.html

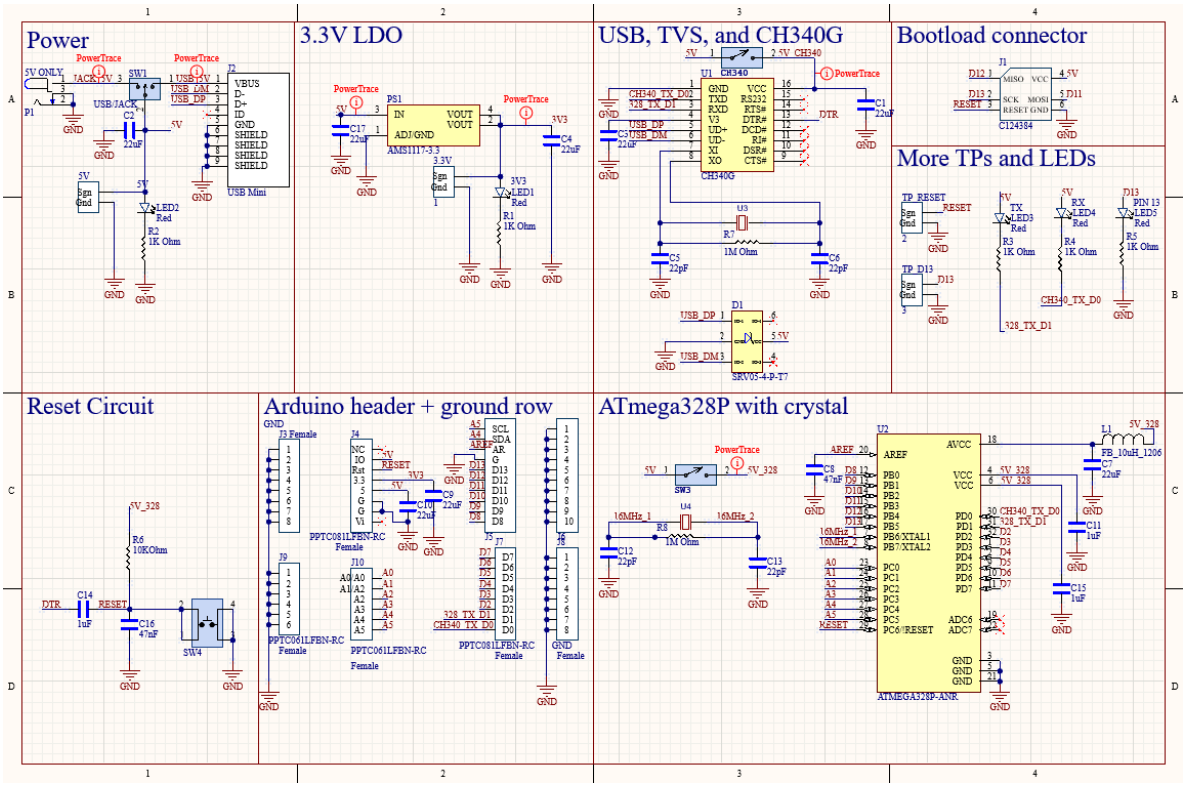
Differences from the commercial board (Whadda ATmega328 UNO development board):

1. **Second row of ground header pins.** Each signal leaving the board has a return pin right beside it instead of sharing the one ground pin on the Whadda digital header, which keeps the return loops small.
2. **A decoupling capacitor at every VCC pin, with AVCC filtered separately.** The switching current comes from the capacitors at the pins instead of through the power path on the board.
3. **No copper pour on the signal layer, and a ground plane only broken by a few small cross-unders.** The commercial board has a copper pour on both layers. On the top layer, the pour is cut into small pieces by traces and pads, and those pieces couple noise between nearby nets and can act as small antennas. My board does not have a pour on top and uses the whole bottom layer as one continuous ground plane, so each return current flows directly under its trace.

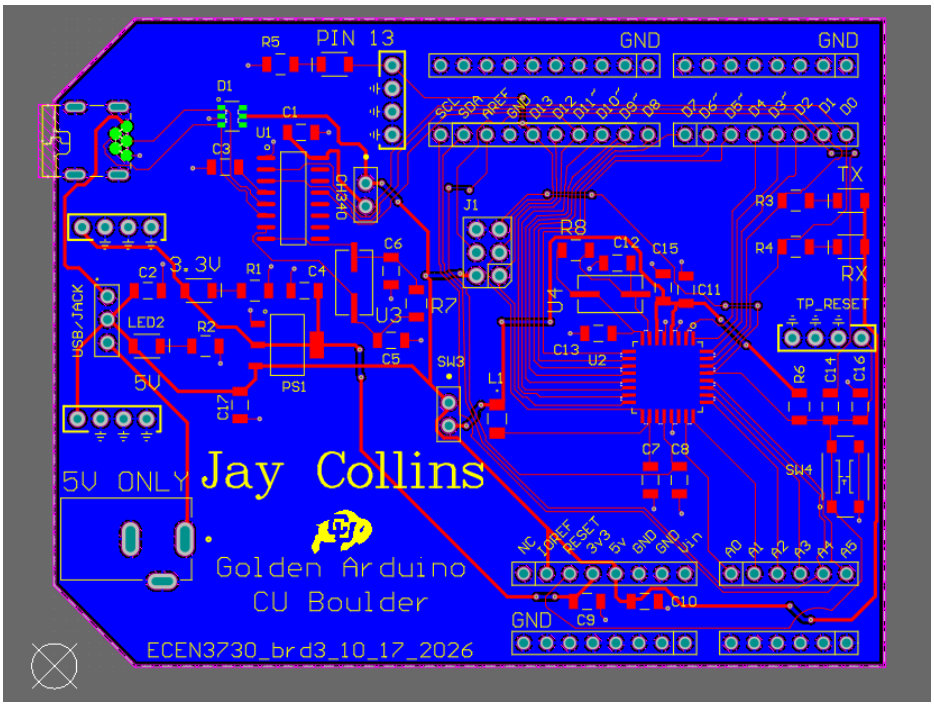
Block Diagram:



Schematic:



Layout:



Requirements to "work":

- The 5V and 3.3V rails measure about 5V and 3.3V from either power source, and the indicator LEDs turn on.
- The 328 accepts the bootloader, the computer sees the CH340G as a COM port, and a modified Blink sketch toggles pin 13.
- The noise shield plugs into the headers and fits.
- With the noise shield, the switching noise is 20% to 50% of the commercial board and the near field emissions are less than 10% of the commercial board.

Risk Reduction:

- I validated the schematic and ran the design rule check on the layout, and checked each footprint against its datasheet before ordering.
- I named the TX and RX nets after the chip that drives them (CH340_TX_D0 and 328_TX_D1) so the TX to RX connection was easy to check.
- Isolation switches let me turn the CH340G or the 328 off on its own, and the indicator LEDs and test points will show that certain parts of the board are operating correctly.

- The switch makes it impossible to power the board from USB and the jack at the same time.
- I brought the board up in order: power rails first, then the bootloader, then Blink, and only then the noise measurements.
- I took every noise measurement the same way on both boards, with the spring tip on the 10x probe, and measured the rise and fall times so the comparison was fair.

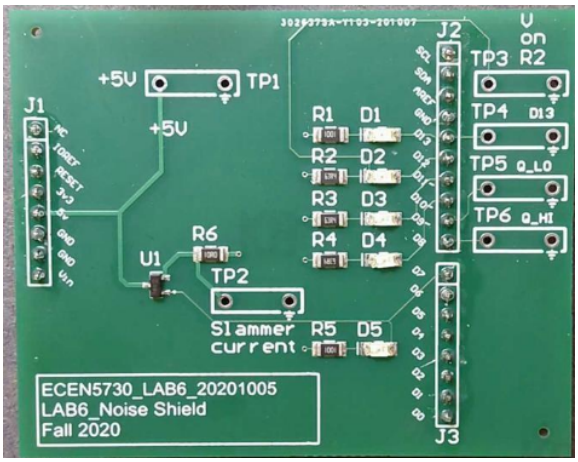
Implementation:



The assembly process was very tedious, especially the small pins on the ICs. Once it was all assembled, I took basic measurements relating to voltage and verified that it was around 3.3 V and around 5 V where expected. Then, I successfully burned the bootloader and turned my board into an Arduino Uno. This process went smoothly. I then uploaded a modified Blink sketch over USB through the CH340G, with the delay changed so the pin 13 LED toggled at 5 Hz, which showed that both the 328 and the CH340G were working.

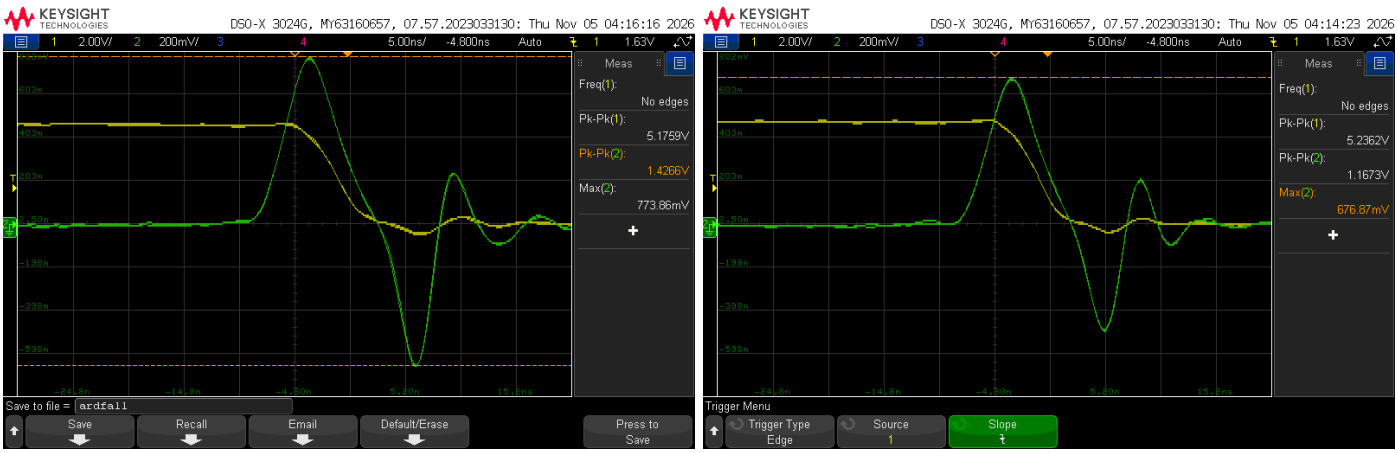
Scope Measurements and Calculations:

All of these measurements used a switching noise shield pictured below. Each one was taken the same way on the commercial Arduino board and on my board.

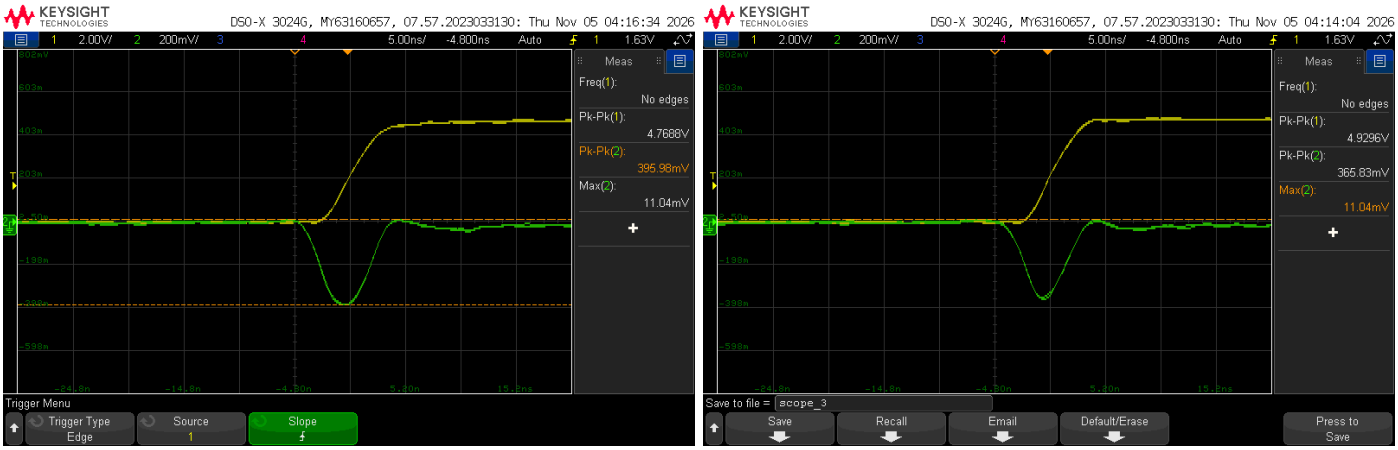


Quiet low ground bounce (pins 10, 11 and 12 switching):

The first measurement is the quiet low (D9) while pins 10, 11 and 12 switch at the same time. The scope was triggered on pin 13, which switches on the same instruction. Channel 1 is the switching edge at 2 V/div and channel 2 is the quiet low at 200 mV/div.



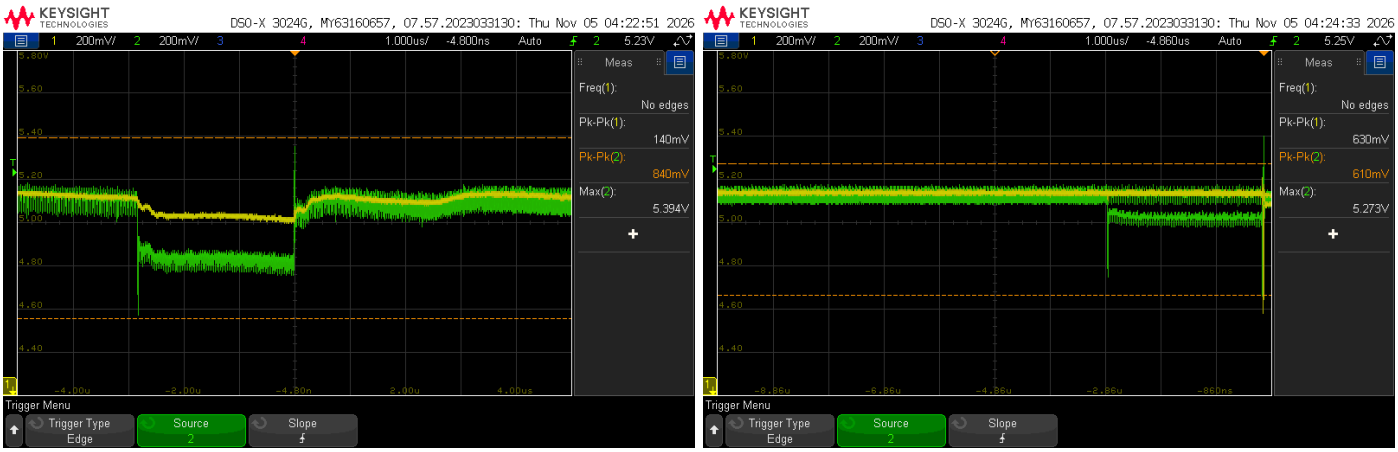
On the falling edge, the commercial board (left) rang to 1.43 V peak to peak and my board (right) rang to 1.17 V peak to peak.



On the rising edge, the commercial board (left) dipped 396 mV and my board (right) dipped 366 mV. My board had about 82% of the commercial board's quiet low noise on the falling edge and 92% on the rising edge. This noise difference comes from the return path on the board, which my continuous ground plane and ground row shortened.

Quiet high and 5V rail when the chip is the aggressor:

Next I measured the 5V rail on the board (channel 1) and the quiet high, D8 (channel 2), while the I/O pins switched, both at 200 mV/div.

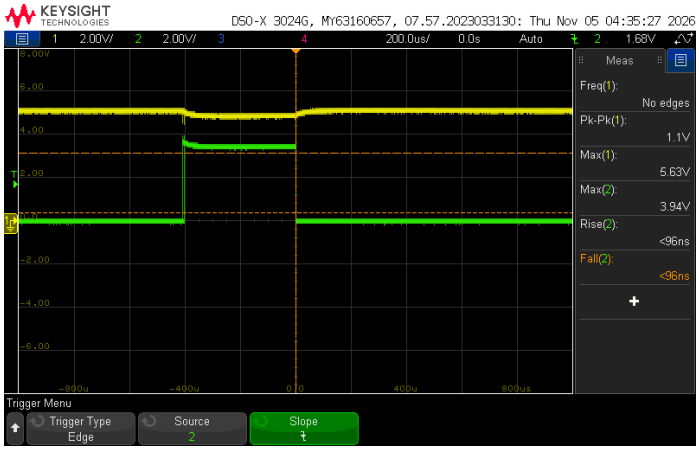


On the commercial board (left) the quiet high dropped about 280 mV for the length of the pulse and the board's 5V rail drooped about 100 mV with it. On my board the quiet high dropped only about 80 mV and the 5V rail showed no droop I could see on this scale. The peak to peak values on screen are mostly the short spikes at the edges, so I compared the steady droop instead. My board had about 30% of the commercial board's quiet high droop. This is where the decoupling capacitor at

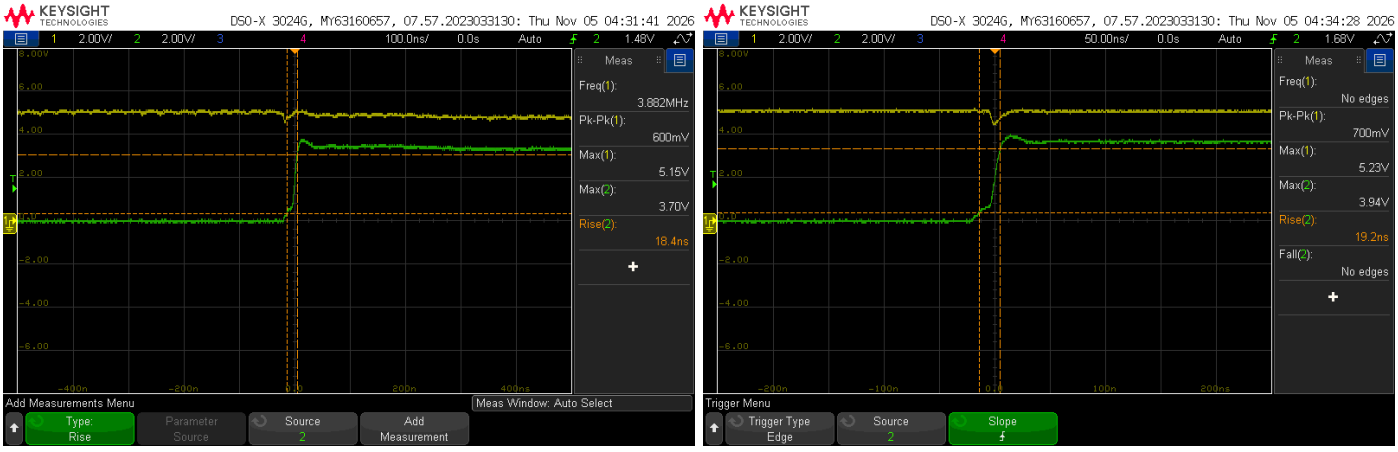
each VCC pin and the separate AVCC filter show up, since the switching current comes from the capacitors right at the pins instead of through the board's power path.

5V rail and quiet high when the board is the aggressor (slammer):

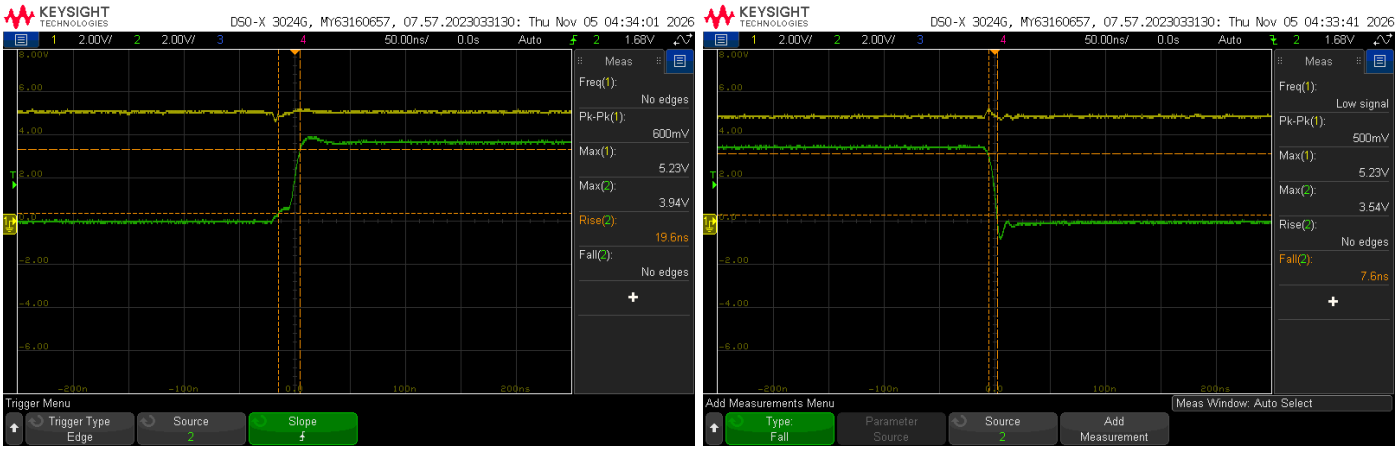
For this measurement pin 7 drove the slammer MOSFET on the shield, which pulls current from the 5V rail through a 10 Ω resistor. Channel 2 is the voltage across the 10 Ω resistor, so 1 V on channel 2 is 100 mA, and the scope was triggered on channel 2.



On my board the slammer drew 3.94 V / 10 Ω = 394 mA for about 400 µs, and the commercial board drew 3.70 V / 10 Ω = 370 mA. During the pulse, my 5V rail dropped about 0.2 V, which gives an output Thevenin resistance of about 0.2 V / 0.394 A = 0.5 Ω for the 5V rail.



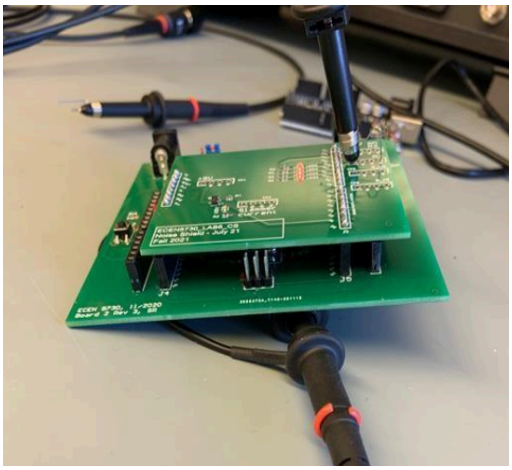
On the rising edge of the current, the 5V rail had 600 mV of noise peak to peak on the commercial board (left) and 700 mV on my board (right). The rise times are also both about 19 ns, meaning the noise comparisons are valid since the dt is the same for both.

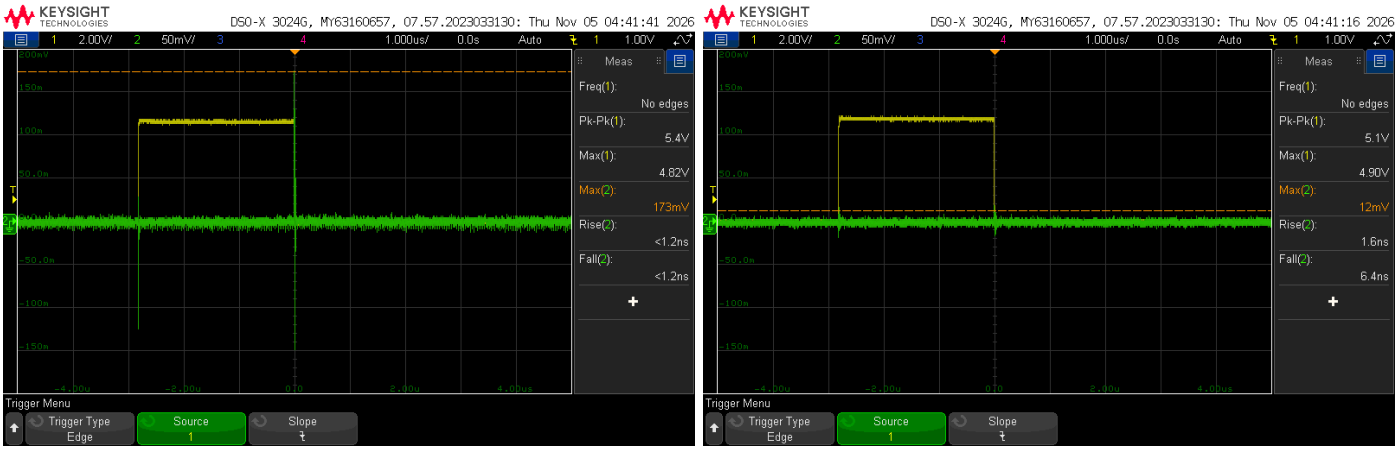


On my board, the quiet high had 600 mV of noise peak to peak on the rising edge and 500 mV on the falling edge. I did not capture the quiet high on the commercial board, but the 5V rail noise during the slammer pulse was about the same on both boards, so I would expect the quiet high to be similar. My rail noise was slightly higher, but my board also drew more current. This noise is set by the inductance between the MOSFET drain and the nearest decoupling capacitor, and most of that path is on the shield and through the header pins, which are the same for both boards. My 22 uF capacitor at the 5V header pin helps, but it cannot fix the inductance on the shield.

Near field emissions:

For the last measurement I used a second 10x probe with its tip shorted to its ground as a pickup loop. I moved it around under each board while the scope was triggered on pin 13 and kept the position with the most pickup, which was under pin 13. Setup method shown below.





The commercial board's peak pickup was 173 mV. My board's peak was 12 mV, which is about 7% of the commercial board. This was the biggest improvement on the board. The commercial board's return currents spread out and form large loops that radiate. My continuous ground plane keeps each return current directly under its signal trace, so the loops are small and very little field reaches the probe.

Summary:

My targets were 20% to 50% of the commercial board's switching noise and less than 10% of its near field emissions. The near field emissions met the target at 7%. The quiet high droop met it at about 30%, and the 5V rail stayed flat when the chip was switching. The quiet low ground bounce improved only 8 to 18%, and the slammer noise was about the same. Both of those are set mostly by things the board cannot change, the package ground inductance and the slammer path on the shield. The things the board does control, the decoupling at each pin and the full ground plane, made the largest difference.

Project Analysis:

Practices that worked well:

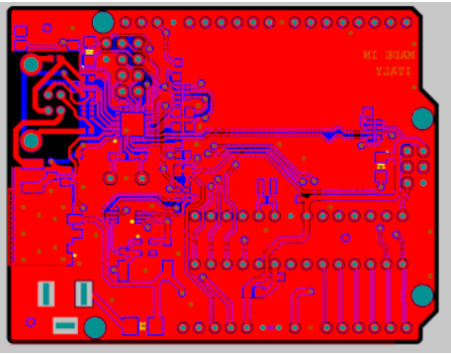
- Labeling the individual components on the board made it easy to assemble.
- Labeling the test points helped avoid any confusion when testing.
- Writing the exact file name of the board made it easy to go back in and modify later.
- Proper placement of the decoupling capacitor reduced noise, as well as the inclusion of a ground plane, without a pour on the signal layer.

- The use of indicator LEDs, test points, and switches made it very easy to obtain accurate measurements.
- Taking scope captures using the same scale for both boards made comparisons easy.

What I would do differently if done again:

- In the future, I will use a soldering iron with a finer tip to make it easier to accurately solder smaller components.
- In the layout, I will wire the more complex parts first so that I can use fewer cross-unders to correct initial routing.
- I would have liked to use a bigger TVS because that was the hardest thing to solder on.

Conclusion:



This was the most complex PCB I have designed and assembled to date. It worked without any debugging which was very satisfying. Using all of the best practices in order to make an Arduino board without nearly as much noise was a great learning experience. When comparing my board to a commercial Arduino Uno, it is very obvious why there is so much noise on theirs, for example, both layers having a copper pour (shown to the right). In their defense, these boards are rarely being used in cases that this noise would affect much. Overall, this was one of my favorite projects I have done in college and I will certainly be sure to implement the design practices (such as close decoupling capacitors and no copper pour on signal layers) in the future.